

Universität Stuttgart

Signal processing and Analysis of human brain potentials (EEG)

Project Report

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1. Introduction and task description

Our team OneTwoCats (Mila Bochkova & Didem Celik) have worked on replicating the results from the study “Task-level value affects trial-level reward processing” by C. D. Hassall, L. T. Hunt & C. B. Holroyd (2022). The authors of the study investigate how the overall value of a task influences how we process individual trial outcomes. Hassall et al. (2022) focus on the role of the anterior cingulate cortex (ACC) in reward processing by looking at a neural signal called the reward positivity (RewP), which reflects reward-related outcome evaluation. The source code of our preprocessing pipeline and replication results is available on GitHub (<https://github.com/didmcelik/reward-erp-eeeg-semester-project>).

1.1 Goal of the project

The goal of our project was to replicate and extend the findings of Hassall et al. (2022) using the data of twelve participants that is publicly available (<https://openneuro.org/datasets/ds004147/versions/1.0.2>). Specifically, we aimed to replicate the main result of the paper – task-level value modulates the RewP amplitude in feedback-locked EEG –, and to demonstrate that this effect is robust to an independent preprocessing pipeline, which we implemented in Python (using MNE). The source code of our pipeline can be found here: <https://github.com/didmcelik/reward-erp-eeeg-semester-project>

1.2 Overview of the structure

The report is organised as follows. Following this introduction, Section 2 contains the summary of the Original Paper, including the main contribution, experimental design, the research question, and analytical approach used by Hassall et al. (2022). Section 3 describes how we replicated the study, which involves our preprocessing pipeline, the implementation differences, and outcomes. The third section discusses the results, comparing our results with the authors’ findings. The final section includes additional analysis beyond the original ERP analysis, such as decoding.

2. Original Paper

2.1 Main contribution

Hassall et al. (2022) provide evidence that the average task value influences trial-level reward processing in the ACC. Using EEG, the authors show that the amplitude of the RewP component following reward outcomes is significantly reduced in high-value tasks compared to mid-value tasks, even when the specific reward magnitude and outcome expectancy of the trials are identical. This suggests that the ACC tracks relevant task variables to regulate reward-related neural activity.

2.2 Experiment design

The authors used a probabilistic reinforcement learning task framed as gambling in three different “casinos”. EEG was recorded continuously while participants completed the task.

In each task, participants made choices that led to outcomes (wins or losses). Each casino represented a different average task value:

1. Low-value: all cues resulted in a 50% probability of a win;
2. High-value: all cues had one high-reward arm (80%) and one low-reward arm (20%), allowing participants to learn the optimal choice;
3. Mid-value task: an even split of 50% and 80/20% cues.

Each casino had six unique coloured shapes representing “slot machines”. Within each casino, participants encountered each of the six slot machines 24 times in random order, resulting in 144 trials per casino. On each trial, participants selected one of two “arms” by pressing a keyboard key. After a 1000 ms delay, which separated cue-related and response-related activity, feedback was presented, indicating a win (\$0.03) or a loss (\$0.00). While individual outcomes were identical across tasks, the overall reward rate differed, allowing the authors to test whether task-level value influences trial-level reward processing.

2.3 Research question

The primary research question was: “How does average task value affect reward-related ACC activity?”. Hassall et al. (2022) looked to understand if trial-level reward processing measured via RewP interacts with task-level value processing, or if rewards are evaluated purely as “one-step” computations, independent of the broader context.

2.4 Author’s preprocessing pipeline

The authors used a preprocessing pipeline implemented in MATLAB using the EEGLAB library. The EEG data was recorded from 35 electrode locations and downsampled to 250 Hz. The main preprocessing steps included:

1. Filtering: a bandpass filter (0.1 – 30 Hz) was applied, along with a notch filter for line noise (60 Hz for participant data from UVic site, 50 Hz for Oxford site);
2. Re-referencing: data was re-referenced to the average of the mastoid signals;
3. Bad channel handling: noisy channels were identified through visual inspection and removed;
4. Artifact removal: Independent Component Analysis (ICA) was performed using the *runica* algorithm. To improve decomposition quality, ICA was trained on extended data segments. Ocular components were identified by the *ICLabel* function and removed before reconstructing the cleaned EEG signal;
5. Epoching: epochs were constructed from -200 to 600 ms around cues and feedback;
6. Artifact rejection: epochs were rejected if they contained a voltage change exceeding 40 μ V per sample or a total change of 150 μ V across the epoch;
7. Trial exclusion: the first ten trials of each task were excluded to account for the learning/familiarization phase of the participants.

2.5 Author’s analysis

Hassall et al. (2022) used a combination of:

1. ERP analysis: the RewP was quantified using a difference wave method (mean win ERP minus mean loss ERP). The signal was measured at electrode FCz within a 240 – 340 ms time window;
2. Statistical testing:
 - a. One-way repeated-measures ANOVA testing confirmed the task value manipulation by comparing win proportions;
 - b. Repeated-measures t-tests were used to compare RewP scores across tasks, specifically matching for outcome frequency to avoid confounds;
 - c. Correlation analysis related participant performance levels to the magnitude of the RewP reduction;
3. Computational modeling: the authors fit a reinforcement learning model to the behavioral data using the *fmincon* function, in order to estimate the learning rate and temperature parameters for each participant and task.

3. Our replication

Only the dataset from the University of Oxford site was publicly available. As a result, our replication included data from 12 participants, compared to 36 participants in the original study by Hassall et al. (2022). The smaller sample size may affect the robustness of statistical comparisons, however, it still allows us to test whether the core effect of modulation of the RewP by task-level value, can be reproduced.

3.1 Research question

The main research question of our replication was: “Does the average task value (how “rich” a task is on average, in terms of reward) affect how the brain processes individual reward outcomes (RewP)?”.

More specifically, we aimed to determine whether RewP amplitude differs across low-, mid-, and high-value task contexts, even when outcome properties are matched.

3.2 Technical Implementation

All analyses are implemented in Python using the MNE-Python package. The preprocessing and analysis pipeline was split into separate scripts for each of the pipeline steps, to ensure reproducibility and ease of debugging. Automated methods were selected to reduce subjective decision-making. Sanity checks were performed at each stage, including visual inspection of raw data before and after filtering, verification of event timing, inspection of ICA components and topographies, ERP waveform visualization per condition, and trial count verification after exclusions.

3.3 Our preprocessing pipeline

We aimed to replicate the original processing pipeline while implementing several modifications. The comparison can be seen below (Table 1).

Step	Authors (Hassall et al., 2020)	OneTwoCats
Bad Channels	Manual identification & removal (≤ 4 channels)	Automated detection using statistical metrics
Filter	0.1 – 30 Hz	0.1 – 40 Hz
50 Hz	Notch	Zapline
Re-referencing	To mastoids (site 1: M1, M2; site 2: TP9, TP10)	To mastoids (TP9, TP10)
Artefacts	ICA (<i>runica</i>) + ICLabel EEG LAB (EOG removal)	ICA (<i>picard</i>) + IC Labeling (MNE-ICLabel) + EOG detection
Epoching	-0.2 – 0.6 s (feedback)	-0.2 – 0.6 s (feedback)
Baseline	-200 – 0 ms	-200 – 0 ms
Excluded trials	Invalid + learning (first 10 trials)	Invalid + learning (first 5 trials)
Averaging	Mean ERP	Mean ERP
Statistics	one-way repeated-measures ANOVA; t-tests at FCz	Cluster-based permutation tests

Table 1: Difference between authors' and our pipeline.

1. Bad Channel Handling:

The original study used manual identification and removal of channels of up to 4 channels (if more bad channels were found, the participant was excluded from further analysis). Our pipeline uses automated detection using three statistical metrics: standard deviation (overall variance), peak-to-peak amplitude (extreme values), and kurtosis (distribution shape). For each metric, channels with z-scores exceeding 3σ (3 standard deviations from the mean) were flagged as bad. Identified channels were interpolated in order to preserve spatial information. Automated detection was chosen since it reduces subjective bias and saves time (MNE Forum, n.d.).

2. Filtering

We decided to use a slightly higher low-pass cutoff (0.1 – 40 Hz) compared to authors (0.1 – 30 Hz) in order to preserve additional high-frequency information.

3. Line Noise Removal (50 Hz):

While Hassall et al. (2022) used a notch filter at 50 Hz, we have chosen the Zapline algorithm. Traditional notch filters attenuate the entire frequency band around 50 Hz, potentially affecting neighbouring signal components. Zapline removes just the power line

signal while preserving surrounding spectral information, reducing waveform distortion (MNE Forum, 2021).

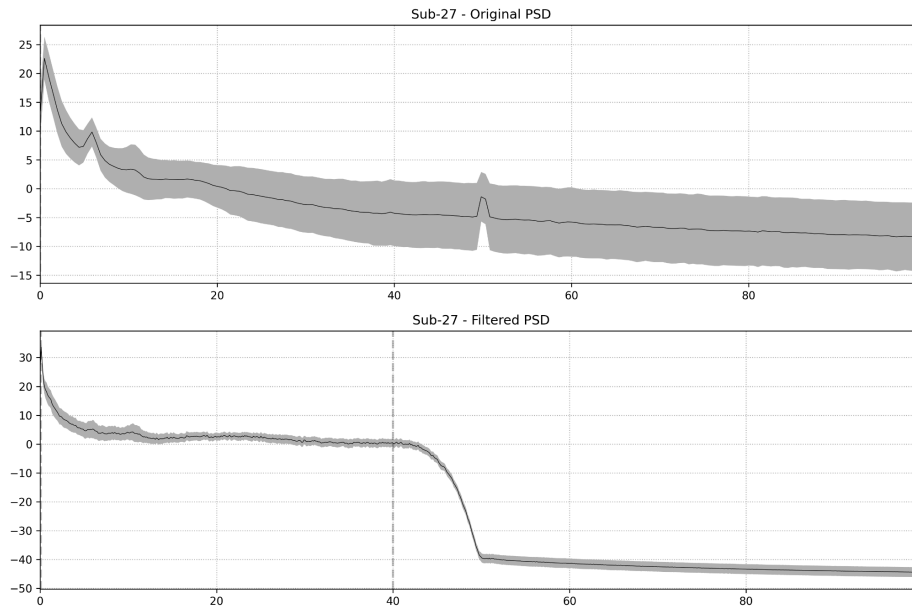


Figure 1: PSD before and after filtering (subject 27)

4. Re-referencing

Same as the authors. We use the average of mastoid electrodes (TP9, TP10 at Oxford site). Common Average Reference is generally recommended for high-density (>64 channel) systems (Brain Products, n.d.). Since this dataset uses 32 channels, mastoid referencing was selected to match the original design.

5. Artifact Removal

The original study used *runica* algorithm & ICLabel for ocular component rejection. We used *Picard* ICA algorithm, along with MNE-ICLabel. *Picard* is computationally faster and converges reliably while producing components comparable to *runica*, and automated labeling minimizes manual component selection (Frank et al., 2022). We also set channels Fp1 and Fp2 designated as EOG, which are used for eye movement artifact detection.

6. Epoching, Baseline Correction, and Averaging

Remains the same between our pipeline and authors in order to ensure direct comparability of ERP results.

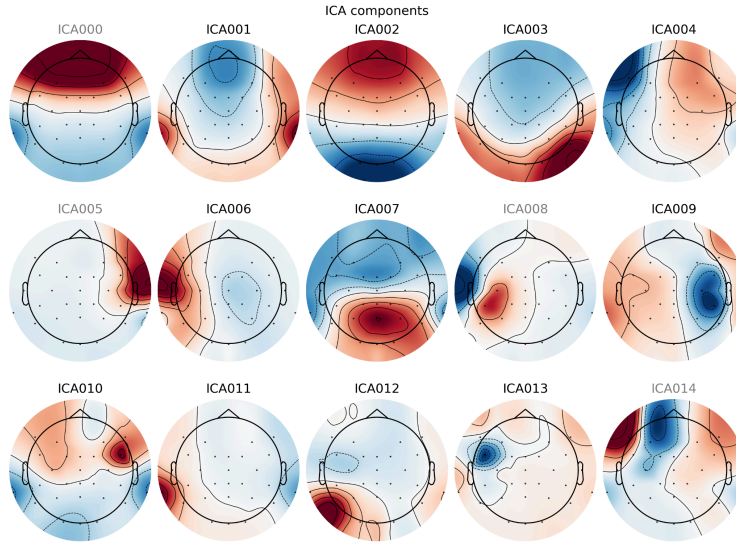


Figure 2: Identified ICA components (subject 35)

7. Trial Exclusion

The original study excluded the first 10 trials per task, as well as invalid trials. We decided to drop fewer trials in the learning phase (5 instead of 10) to preserve statistical power in the smaller sample while still accounting for early learning effects. This allows us to keep more usable data without altering the learning phase.

8. Statistics

For statistics, we have chosen cluster-based permutation testing to control for multiple comparisons without relying on a priori selection of a single electrode or time window (Sassenhagen & Draschkow, 2019).

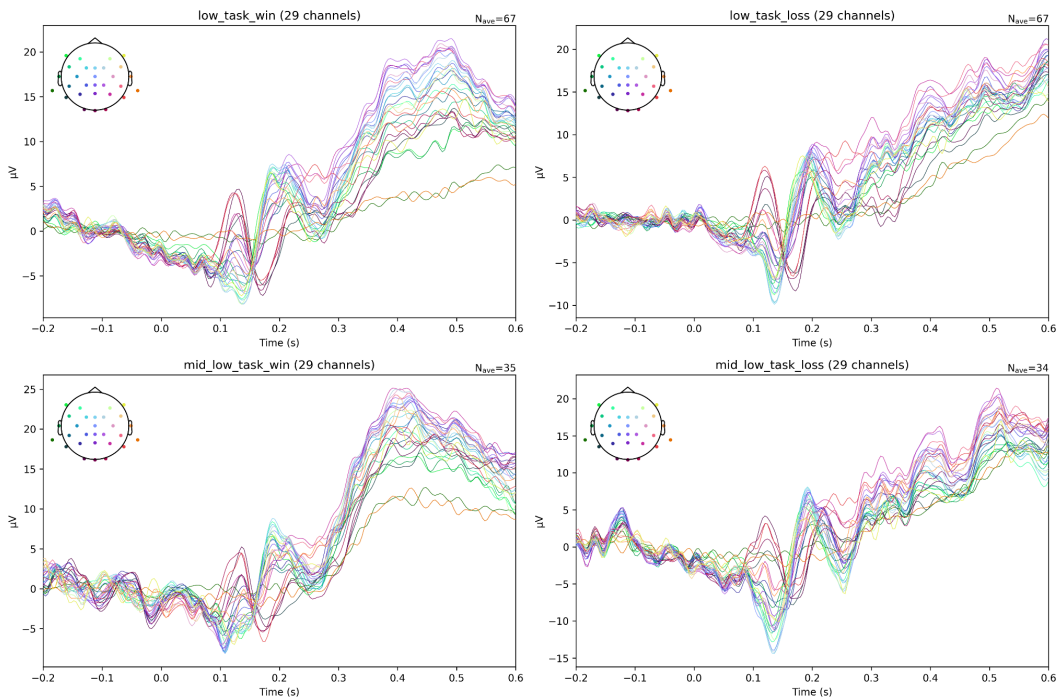


Figure 3: Overview of epochs (subject 34)

3.4 Our analysis

Following preprocessing, we computed feedback-locked ERPs separately for win and loss trials within each of the four task-cue combinations (low, mid-low, mid-high, high). The event mapping is:

- Stimulus/S 2 : Low Task, Low Cue
- Stimulus/S 12 : Mid Task, Low Cue (Mid-Low)
- Stimulus/S 22 : Mid Task, High Cue (Mid-High)
- Stimulus/S 32 : High Task, High Cue

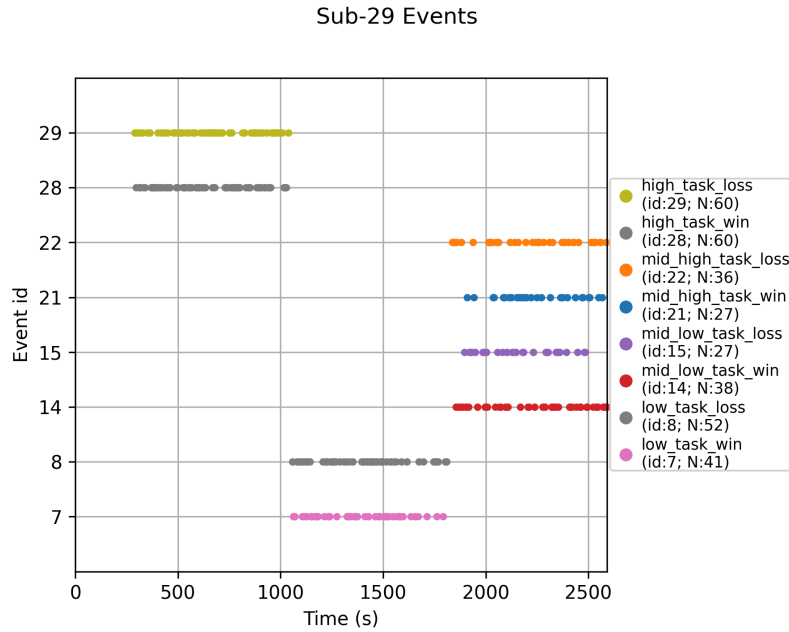


Figure 4: Overview of events (subject 29)

The RewP was calculated as the difference wave (win minus loss) within each condition. The RewP amplitude was quantified as both the peak (maximum) and mean amplitude within the 240 – 340 ms time window at electrode FCz. This allowed for a direct comparison with the reported results of Hassall et al. (2022).

To test whether the RewP differed significantly between win and loss outcomes, we conducted cluster-based permutation tests using the MNE-Python implementation. For each of the four expectancy conditions, we performed a paired one-sample permutation test on the win-loss difference data across electrodes and time points (-200 – 600 ms), using 1,000 permutations and automatic threshold selection. The test identified contiguous clusters of electrodes and time points showing consistent differences between win and loss feedback, with statistical significance determined at $p < 0.05$ (cluster-corrected).

To characterize the topographic distribution and temporal dynamics of reward processing, we created scalp topographic maps of the win-loss difference wave at multiple time points (200, 252, 292, 352, and 400 ms) and generated visualizations showing the spatial evolution across the RewP time window. We also examined individual subject waveforms across all eight

conditions (4 expectancy levels $\times$ 2 outcomes) to assess inter-individual variability and confirm the consistency of the RewP effect across participants.

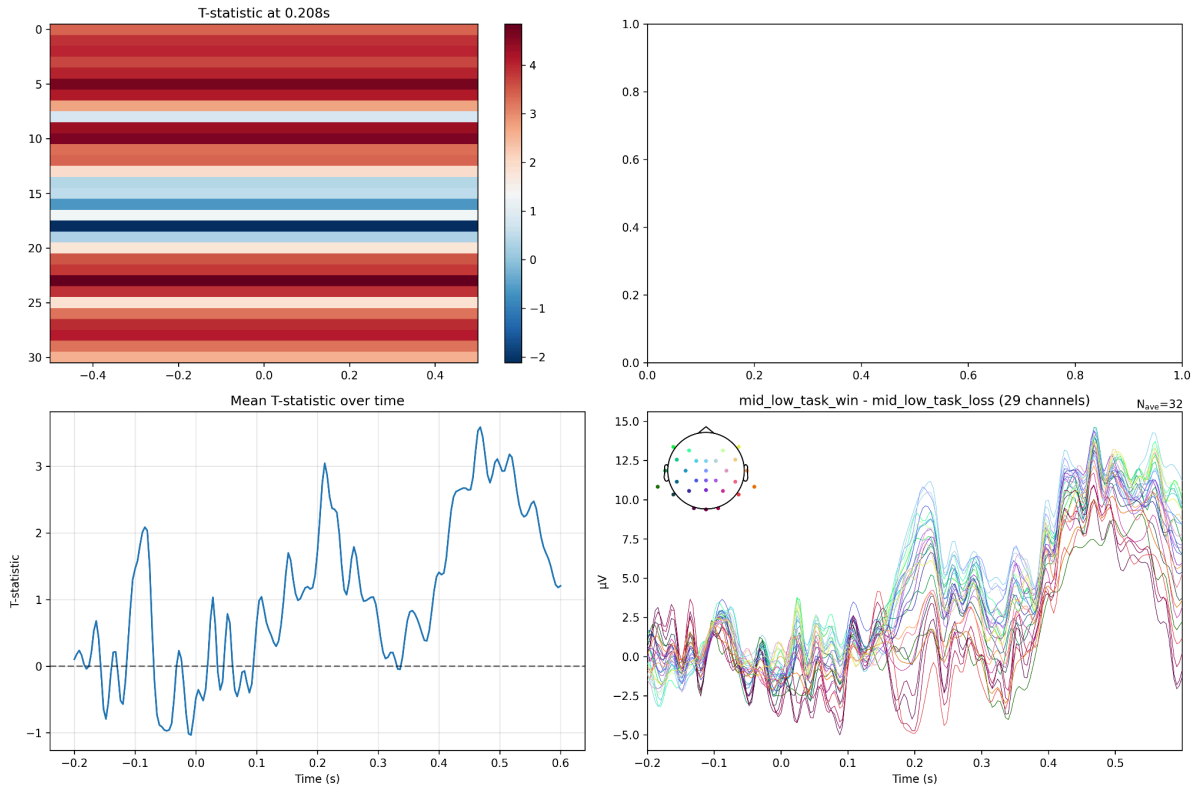


Figure 5: Cluster-based permutation tests results for Mid-Low Win vs Loss (subject 36)

4. Results & Discussion

4.1 Feedback-locked ERPs (Win vs Loss)

The feedback-locked ERP waveforms for win and loss trials in our replication strongly resemble those reported by Hassall et al. (2022). As can be seen in comparison plots (Figure 6), both datasets show a clear fronto-central positive deflection following reward feedback, and a differentiation between win and loss waveforms within the time window (240–340ms).

This similarity suggests that our pipeline successfully preserved the core ERP structure and that the neural signal corresponding to reward processing was reliably recovered. However, while the general form matches authors' results, the difference waveforms (win – loss) show deviations in amplitude shape. In the post-feedback window (>350 ms) our plot shows greater amplitude compared to authors' ($\sim 20\mu\text{V}$ vs $\sim 15\mu\text{V}$).

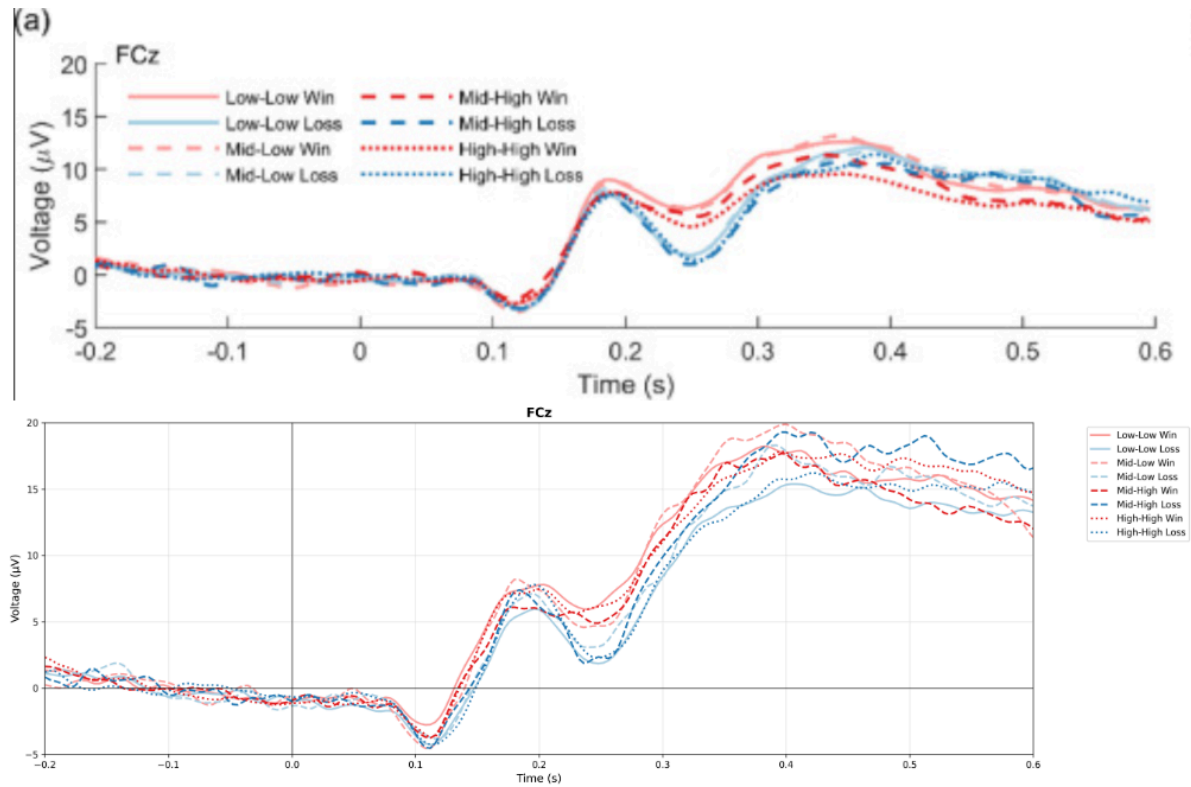


Figure 6: Win/loss waveforms by task value and cue value – all subjects (Hassall et al. – top image, OneTwoCats – bottom image)

Moreover, in the author's results, High-High Win < Mid-High Win consistently throughout the RewP window, but our data shows a pattern where High-High Win starts higher than Mid-High Win (for a period of ~250–300ms), and then becomes lower than Mid-High Win (from ~300ms on). We concluded that this effect is largely due to the smaller sample size. Since the authors had more subjects, it could have led to smoother grand averages with lower peak amplitudes.

4.2 Difference Waveforms

In the original study, the RewP amplitude was reduced in the high-value task relative to the mid-value task (see Figure 7).

In our replication, a positive deflection within the 240 – 340 ms window is present, and the modulation across tasks is visible, but much less pronounced compared to authors. There are several differences between our results compared to difference waveforms by Hassall et al. (2022). Authors have much smoother difference waves with clear RewP peaks in the feedback window. Our data has multiple peaks (~100 – 170ms, ~240 – 270 ms) and is much noisier, which obscures the underlying RewP component.

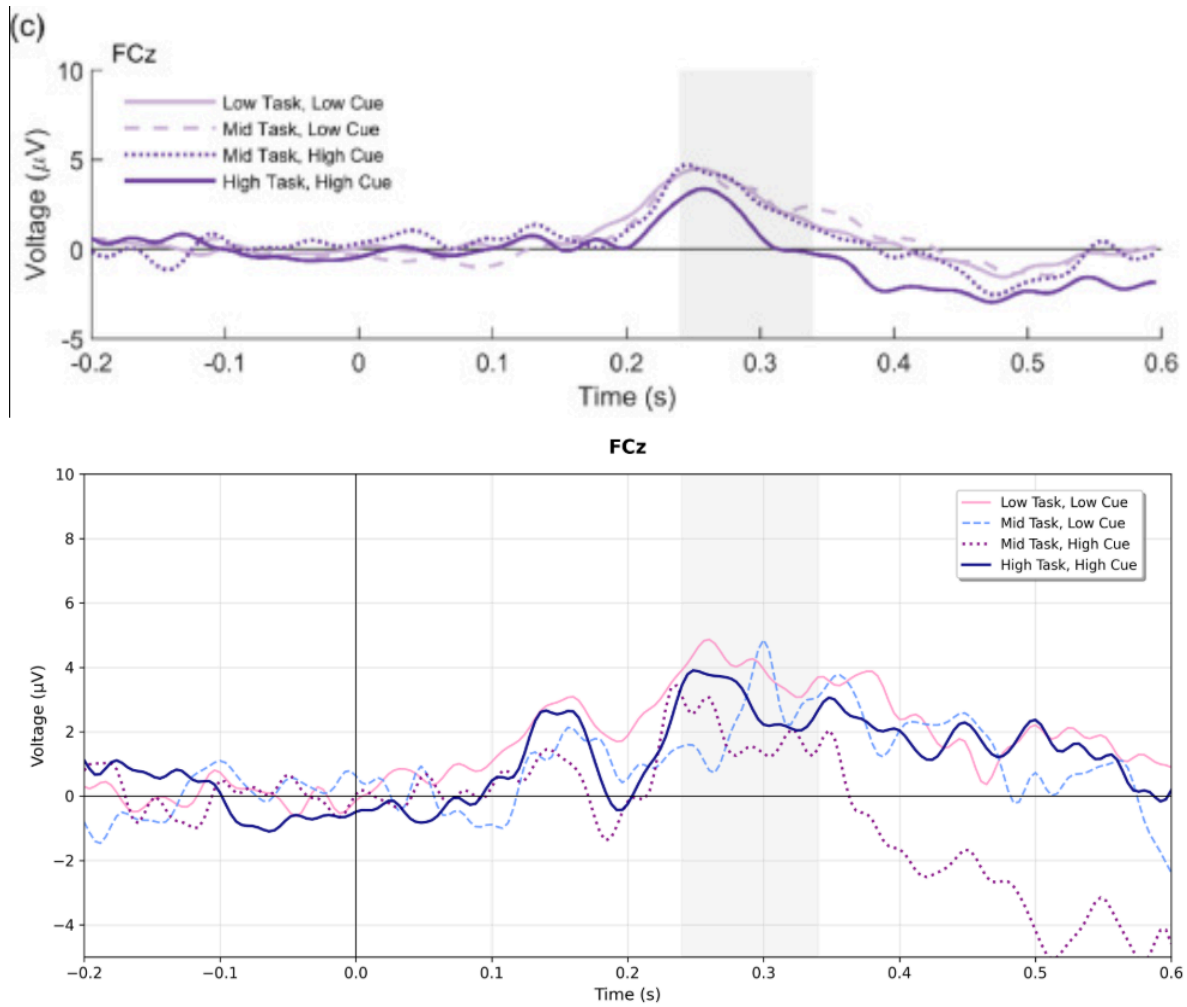


Figure 7: Difference waveforms by task value and cue value (Hassall et al. – top image, OneTwoCats – bottom image)

The most likely cause for this difference is, once again, too few subjects, which makes the individual noise dominate. Besides, we have observed that most subjects have a small number of trials for mid-low and mid-high conditions (e.g. subject 30 has noticeably fewer trials for mid-low and mid-high conditions, compared to the high condition – refer to Figure 8) – this is the case for all subjects except subject 29.

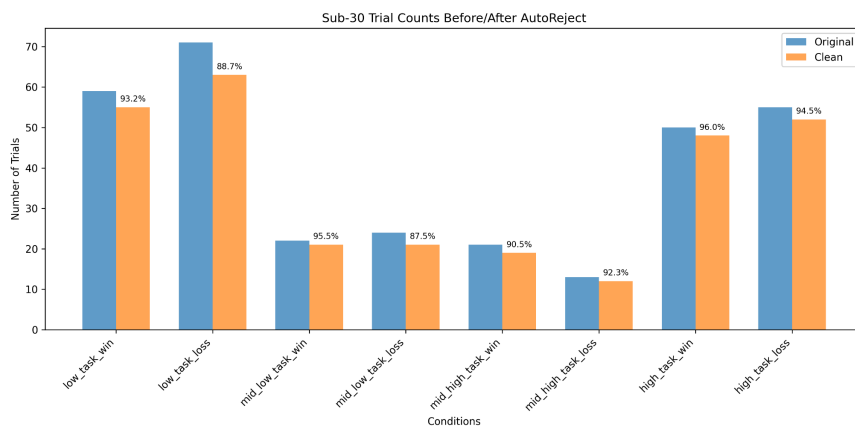


Figure 8: Trial counts comparison for subject 30

4.3 RewP scores by task value

The RewP amplitude comparison across low-low, mid-low, mid-high, and high-high -value tasks shows correspondence with the original findings (Figure 9). In Hassall et al. (2022), the key finding was reduced RewP amplitude in the high-value task relative to the mid-value task.

In our data, the expected ordering trend (mid > high) appears similar, however, variability across conditions is larger. Thus, the pattern is qualitatively consistent but quantitatively weaker.

4.4 Scalp Topography

The scalp distribution of the win-loss difference collapsed across conditions (Figure 10) shows a fronto-central maximum in both datasets, and similar spatial distribution between replication and the original.

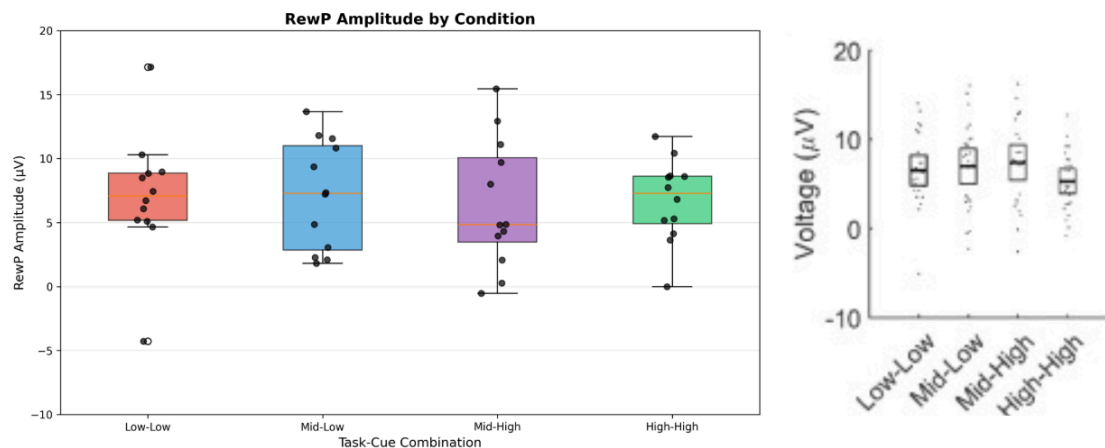


Figure 9: RewP scores by task value (OneTwoCats – left image, Hassall et al. – right image)

The topographical distribution of the RewP is highly consistent with the authors' findings. This supports the interpretation that the same neural process (ACC-related reward processing) was captured in our replication.

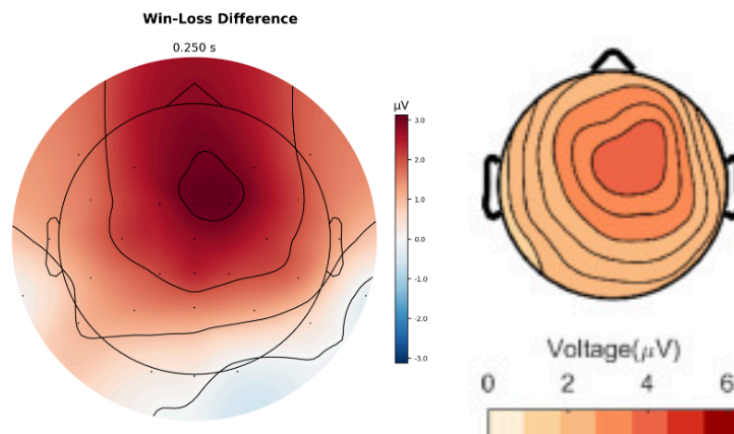


Figure 10: Scalp Topography at 250 ms (OneTwoCats – left image, Hassall et al. – right)

Figure 11 shows the temporal evolution of the win-loss difference wave topography across the scalp at five key time points following feedback onset (200, 252, 292, 352, and 400 ms). Throughout this sequence, the topographic distribution remains consistently frontocentral with slight rightward asymmetry, demonstrating that the reward sensitivity effect is spatially localized to regions associated with anterior cingulate cortex activity and temporally constrained to the 250-350 ms post-feedback window characteristic of the RewP component.

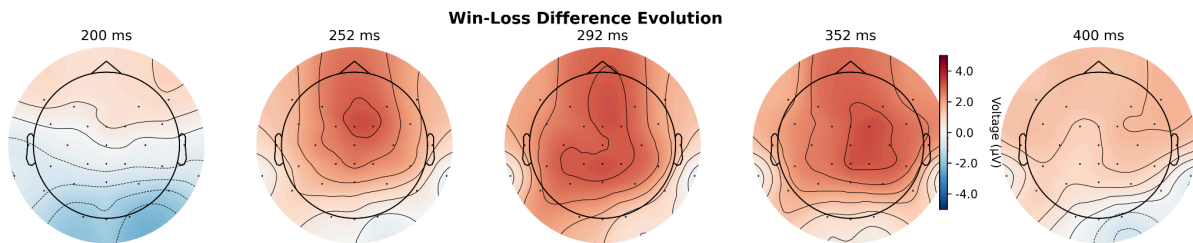


Figure 11: Topography evolution at time points (left to right: 200, 252, 292, 352, 400 ms)

4.5 Discussion

An important aspect of the original study was the exclusion of participants who failed to learn the reward occurrences in the mid- and high-value tasks. Specifically, Hassall et al. (2022) required participants to achieve at least 60% correct responses on learnable trials in both tasks for inclusion in the primary EEG analysis. This resulted in the exclusion of 12 out of 36 participants. In our initial replication, all available participants ($N = 12$) were included in the analysis. Under this approach, the task-level modulation of the RewP was weaker and less clearly differentiated across conditions.

However, when we applied the same learning criterion and restricted the analysis to participants who met the $\geq 60\%$ performance threshold (which resulted in $N = 8$ subjects in our dataset), the pattern of results closely matched that reported in the original study (see Figure 12). In particular, the RewP amplitude for high-value win trials was reduced relative to mid-value win trials, consistent with the hypothesis that average task value attenuates reward-related neural responses.

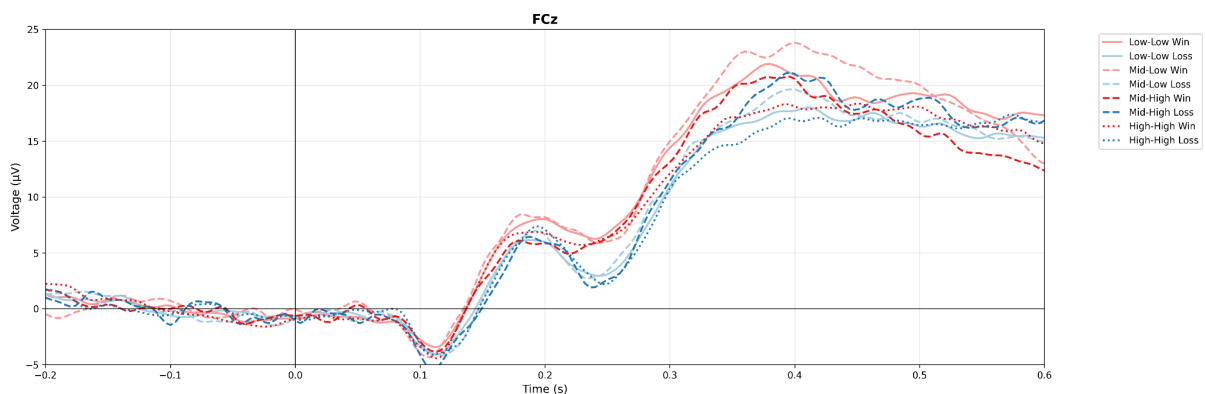


Figure 12: Win/loss waveforms by task value and cue value – learners (subjects 27, 28, 31, 34, 35, 36, 37, 38)

This finding suggests that successful learning of reward occurrences is a necessary condition for observing task-level modulation of the RewP. If participants fail to learn which options are optimal, the task context may not be internally represented as “high-value,” and thus no contextual modulation of reward processing would be expected.

We consider our replication of the Reward Processing study by Hassall et al. (2022) to be partially successful. We have successfully replicated the presence of a clear feedback-locked RewP, frontocentral scalp distribution, win vs loss differentiation, general ERP form, and directional trend of task-level modulation. Nonetheless, the strength and clarity of task-level modulation, as well as smoothness and consistency of difference waves is not fully replicated. We believe that these results come from a smaller number of participants – only data of 12 participants (from the Oxford site) was available to us, compared to 36 participants in the original study. This substantially reduces statistical power, affecting the consistency of difference waves.

To conclude, our replication supports the existence of context-dependent reward processing, but does not fully reproduce the magnitude of the original effect. This outcome is consistent with the expectations, given the reduced sample size.

5. Further Analysis

For further analysis of this dataset, our aim was to answer the following question: “Can EEG activity after feedback tell us whether the trial belonged to a high-value or mid-value task?”. To achieve this, we implemented a classifier to decode task-value from EEG signals.

5.1 Decoding Results

We performed time-resolved multivariate pattern analysis (MVPA) to decode task-value (high vs. mid) from single-trial EEG data. A logistic regression classifier was trained on spatially distributed patterns across all 30 electrodes at each timepoint (–200 to 600 ms relative to feedback onset). Classification accuracy was evaluated using 5-fold stratified cross-validation with balanced classes (N trials per class = 31-64 across subjects, M = 52). Temporal smoothing (50 ms Gaussian kernel) was applied to reduce noise.

Single-subject decoding revealed substantial inter-subject variability (Figure 12, bottom panel). Peak post-feedback accuracy ranged from 52.3% (subject 37) to 69.1% (subject 28), with 7/12 subjects (58%) showing RewP window mean accuracy numerically above chance. This variability likely reflects:

- (1) differences in trial counts after artifact rejection (31-64 trials/class),
- (2) individual differences in neural encoding strength,
- (3) variable signal-to-noise ratios across subjects.

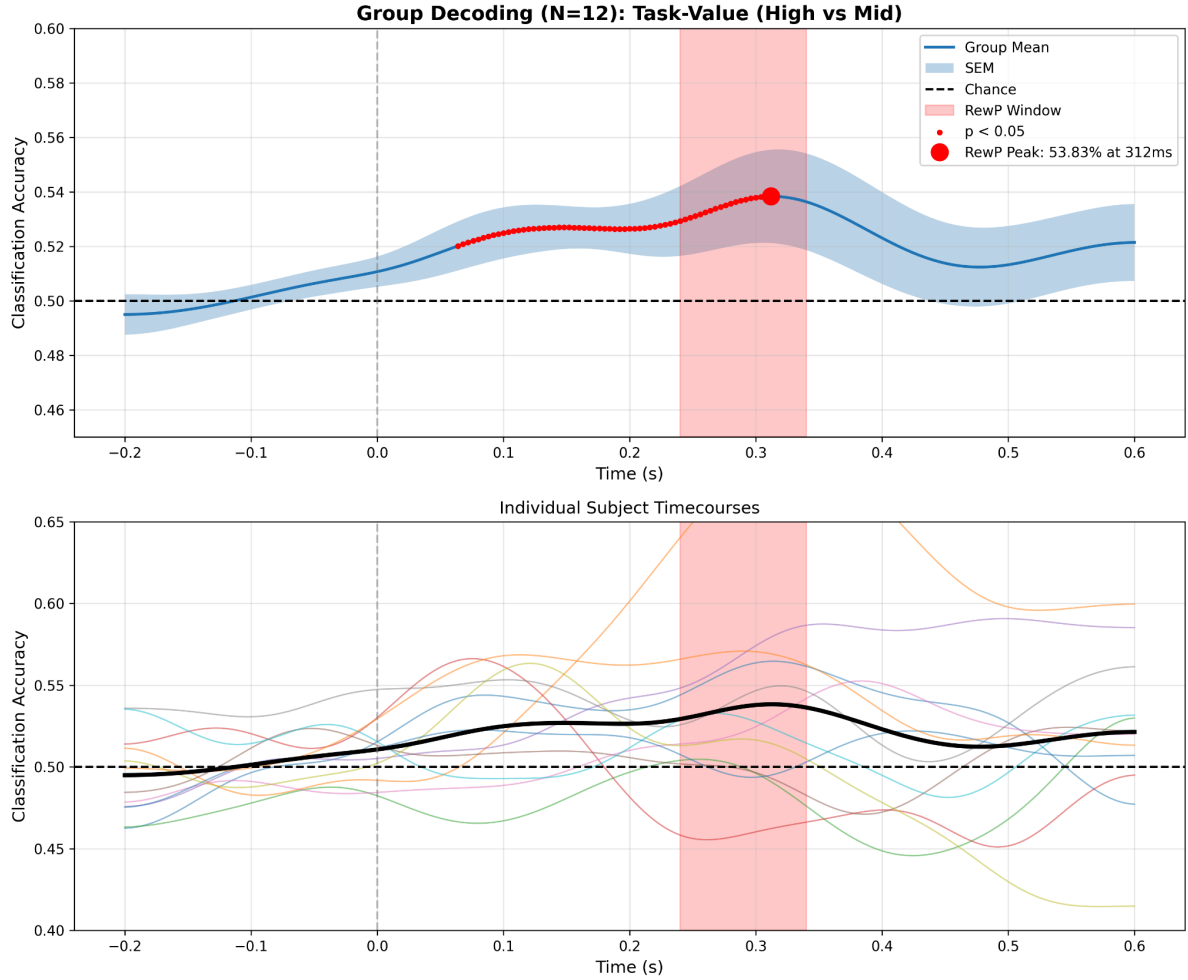


Figure 12: Group-level time-resolved decoding of task value from post-feedback EEG signals

It is worth mentioning that subjects with lower trial counts (e.g., subject 30: $n=31/\text{class}$) showed less stable decoding profiles, with peaks occurring outside the expected RewP window. This is expected due to the effects of limited training data on classifier instability in high-dimensional spaces ($30 \text{ channels} \times 201 \text{ timepoints}$).

Despite individual variability, group-level analysis ($N=12$) revealed above-chance decoding during the post-feedback period (Figure 12, top image). Classification accuracy was significantly elevated from approximately 50 ms to 400 ms post-feedback (64/201 timepoints, $p < 0.05$ FDR-corrected).

Peak decoding accuracy occurred at 312 ms (53.83%, 95% CI [51.2%, 56.5%]), precisely within the RewP time window (240-340 ms). Mean accuracy across the full RewP window was 53.54% ($t(11) = 4.18$, $p = 0.0015$, Cohen's $d = 1.21$), representing a 3.54 percentage point improvement over chance.

These results demonstrate that task-value information is encoded in distributed EEG patterns during feedback processing, validating the functional role of the RewP in value representation.

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During the preparation of this work, the authors used ChatGPT-5 to polish self-authored text. Afterwards, the authors have reviewed and edited the content as needed and take full responsibility for the content of the work.